

JavaScript Full Notes + 30 Problems + Frontend Interview Questions

This PDF is specially prepared for 2+ years frontend developers preparing for interviews. It contains JavaScript notes from very basic to advanced level, 30 coding problems, and most asked frontend interview questions.

1. What is JavaScript?

JavaScript is a programming language used to create dynamic and interactive web applications.

2. Variables

```
let name = "Arun";
const age = 25;
var city = "Delhi";
```

3. Difference between var, let, and const

var is function scoped, let and const are block scoped. const cannot be reassigned.

4. Data Types

String, Number, Boolean, Undefined, Null, Object, Array, Symbol, BigInt.

5. Functions

```
function greet(name) {
  return "Hello " + name;
}
```

6. Arrow Functions

```
const add = (a, b) => a + b;
```

7. Arrays

Arrays store multiple values in a single variable.

8. Array Methods

map, filter, reduce, find, some, every, includes, forEach.

9. Objects

```
const user = {
  name: "Arun",
  age: 25
};
```

10. Destructuring

```
const {name, age} = user;
```

11. Spread and Rest Operator

```
const arr2 = [...arr1];
```

12. DOM Manipulation

getElementById, querySelector, addEventListener, createElement.

13. Event Bubbling and Capturing

Events propagate from child to parent in bubbling and parent to child in capturing.

14. Closures

A closure remembers variables from its outer function even after execution.

15. Hoisting

Variables and functions are moved to the top during execution phase.

16. Scope

Global scope, function scope, and block scope.

17. Callback Functions

Functions passed as arguments to another function.

18. Promises

```
fetch(url)
  .then(res => res.json())
  .then(data => console.log(data));
```

19. Async Await

```
async function getData() {
  const res = await fetch(url);
}
```

20. setTimeout and setInterval

Used for delayed and repeated execution.

21. Debouncing

Limits function calls after user stops typing.

22. Throttling

Limits function execution within intervals.

23. LocalStorage and SessionStorage

Browser storage mechanisms.

24. Event Delegation

Handling child events using parent listener.

25. Call, Apply, Bind

Used to control function context.

26. this Keyword

Refers to execution context.

27. Prototype and Inheritance

JavaScript uses prototype-based inheritance.

28. ES6 Features

Template literals, destructuring, spread, classes, modules.

29. Memory Leak and Optimization

Avoid unnecessary listeners and global variables.

30. Real World Frontend Usage

JavaScript powers React apps, APIs, forms, animations, dashboards, ecommerce apps.

30 JavaScript Practice Problems

- 1 Reverse a string.
- 2 Find palindrome string.
- 3 Find largest number in array.
- 4 Remove duplicates from array.
- 5 Find missing number in array.
- 6 Capitalize first letter of each word.
- 7 Flatten nested arrays.
- 8 Implement debounce function.
- 9 Implement throttle function.
- 10 Create custom map method.
- 11 Create custom filter method.
- 12 Create custom reduce method.
- 13 Count frequency of characters.
- 14 Sort array without sort method.
- 15 Find second largest number.
- 16 Group objects by property.
- 17 Check if two strings are anagrams.
- 18 Create countdown timer.
- 19 Build digital clock.
- 20 Implement infinite scroll logic.
- 21 Create pagination logic.
- 22 Build todo app functionality.
- 23 Create search filter functionality.
- 24 Implement promise chaining.
- 25 Fetch API data using async/await.
- 26 Implement retry API logic.
- 27 Create custom event emitter.
- 28 Implement memoization function.
- 29 Create drag and drop logic.
- 30 Build a simple shopping cart logic.

Most Asked Frontend JavaScript Interview Questions (2+ Years Experience)

- Difference between var, let, and const
- Explain closures with examples

- What is hoisting?
- Difference between null and undefined
- Difference between == and ===
- What is event delegation?
- Explain event bubbling and capturing
- Difference between map, filter, and reduce
- Difference between synchronous and asynchronous code
- Explain promises and async/await
- What is callback hell?
- What is debouncing and throttling?
- Difference between localStorage and sessionStorage
- Explain this keyword
- Difference between call, apply, and bind
- What are pure functions?
- Explain prototype inheritance
- What is memoization?
- How JavaScript execution context works?
- Difference between shallow copy and deep copy
- Explain spread and rest operators
- What are higher-order functions?
- Difference between forEach and map
- What is currying?
- How event loop works?
- What is temporal dead zone?
- What causes memory leaks?
- Difference between function declaration and expression
- What are generators?
- Explain microtasks and macrotasks